Encounters QA Story Report

What this is

> **Context & Scope** Scenario (public-safe)
>
> HarborPoint Health Plan receives weekly encounter batches from delegated vendors for multiple lines of business (e.g., Medicaid and Commercial). Before submitting these encounters to external reporting endpoints (state/federal programs), the Encounters team must catch blocking defects, prioritize remediation, and document a defensible audit trail.
>
> This evidence pack simulates that workflow end-to-end:
>
> - Row-level rejects (missing member/provider fields, invalid dates, eligibility outside coverage, duplicates, etc.)
> - Batch-level flags when operational thresholds are exceeded (duplicate rate >1%, eligibility mismatch >2%, volume shift >15%)
> - Ops decisions: Hold / Reprocess / Submit-with-monitoring
> - UAT traceability for upgrades and vendor feed changes
>
> Out of scope: This is not provider billing QA and not an EDI/837 parser; it's tabular pre-submit QA for payer encounter reporting. Hiring-manager readable summary of a 10-week encounters QA story.

The report is intentionally narrative-first:

- W02 blockers
- W06 eligibility incident + `R902`
- W07 duplicate incident + `R901` + monitor noise
- W10 volume anomaly + `R903` only

All non-incident weeks are shown as `CLEAN` in the storyboard.

Metric Definitions

See Metric Definitions for canonical formulas (numerator/denominator), grain, and anomaly thresholds used in rejects, triage, and tracker outputs.

Explain It Like I'm A Hiring Manager

- **Encounter:** A record of care delivered to a member, used to support plan reporting and payment operations.

- **Batch:** A weekly package of encounter records sent together for preparation, validation, and submission.
- **Reject:** A record-level failure found during validation that must be fixed before clean submission.
- **Batch anomaly flag:** A batch-level warning when rates or volume are abnormal, even if not every record is rejected.
- **Triage summary:** A weekly rollup that shows issue volume by severity so operations teams can decide what to do first.
- **Preparation:** Assembling the encounter batch and checks needed before it can be submitted.
- **Validation and submission:** Running quality checks first, then sending only approved data.
- **Remediation and reprocessing:** Correcting root causes, rerunning the batch, and confirming the fixes hold.
- **Resubmit:** Sending the corrected batch after reprocessing and final review.
- **User Acceptance Testing (UAT):** Documented tests and evidence that prove the workflow and controls work as intended.

Code Legend (Batch Flags)

Code	Name	Threshold	Action
R901	Duplicate-rate anomaly	Duplicate rate > 1% of claims in a batch	Remediate and reprocess before submission
R902	Eligibility-mismatch anomaly	Eligibility mismatch rate > 2% of claims in a batch	Remediate and reprocess before submission
R903	Volume-shift anomaly	Weekly volume shift > 15% vs trailing median	Submit with monitoring controls and follow-up

> **How to interpret this report**
>
> 1. Start with the storyboard to see what happened each week.
> 2. Read the decision column to see operational action (hold, remediate/reprocess, or monitor).
> 3. Use the at-a-glance table for severity totals and top driver codes.
> 4. Use the incident drilldowns and UAT evidence to confirm traceable proof.

Storyboard (10-week timeline)

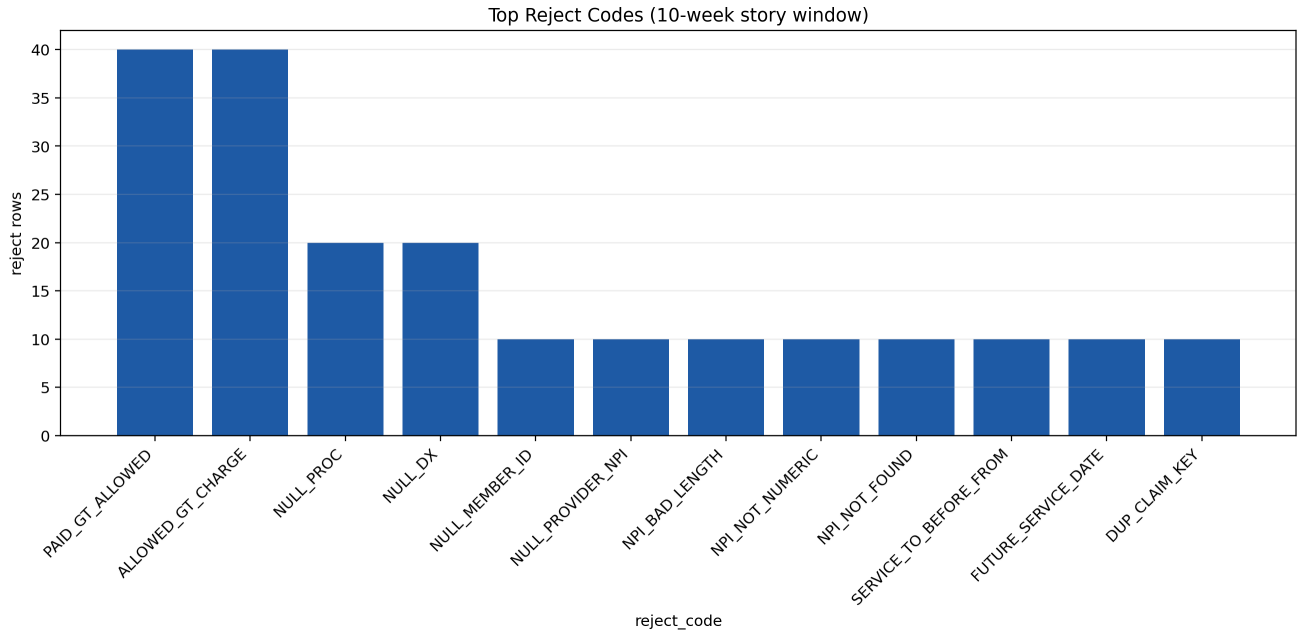
week_start	storyline_label	what_broke	decision	next_week_delta	next_week_label
2026-02-23	CLEAN	CLEAN: no triage rejects in this week.	Proceed; maintain routine monitor controls	+80 rejects	W02 BLOCKERS
2026-03-02	W02 BLOCKERS	BLOCKER=70 from REQUIRED_FIELD (30), PROVIDER (30), DATES (10).	Hold submission (BLOCKER present)	-80 rejects	CLEAN
2026-03-09	CLEAN	CLEAN: no triage rejects in this week.	Proceed; maintain routine monitor controls	0 rejects	CLEAN
2026-03-16	CLEAN	CLEAN: no triage rejects in this week.	Proceed; maintain routine monitor controls	0 rejects	CLEAN
2026-03-23	CLEAN	CLEAN: no triage rejects in this week.	Proceed; maintain routine monitor controls	+7 rejects	W06 R902 ELIGIBILITY-MISMATCH ANOMALY (>2%)
2026-03-30	W06 R902 ELIGIBILITY-MISMATCH ANOMALY (>2%)	HIGH=7 in MEMBER_ELIGIBILITY: R010=6 plus R902 anomaly=1.	Remediate and reprocess before submit	+124 rejects	W07 R901 DUPLICATE-RATE ANOMALY (>1%) + MONITORS
2026-04-06	W07 R901 DUPLICATE-RATE ANOMALY (>1%) + MONITORS	HIGH=11 and MONITOR=120: DUP_CLAIM_KEY=10 + R901=1, FINANCIAL=80, CODE_SET=40.	Remediate duplicates; monitor financial/code-set noise	-131 rejects	CLEAN
2026-04-13	CLEAN	CLEAN: no triage rejects in this week.	Proceed; maintain routine monitor controls	0 rejects	CLEAN
2026-04-20	CLEAN	CLEAN: no triage rejects in this week.	Proceed; maintain routine monitor controls	+1 rejects	W10 R903 VOLUME-SHIFT ANOMALY (>15%)
2026-04-27	W10 R903 VOLUME-SHIFT ANOMALY (>15%)	MONITOR=1 from VOLUME_SHIFT_GT_15PCT (R903) only.	Submit with volume monitor controls	End of window	End of window

Key Findings

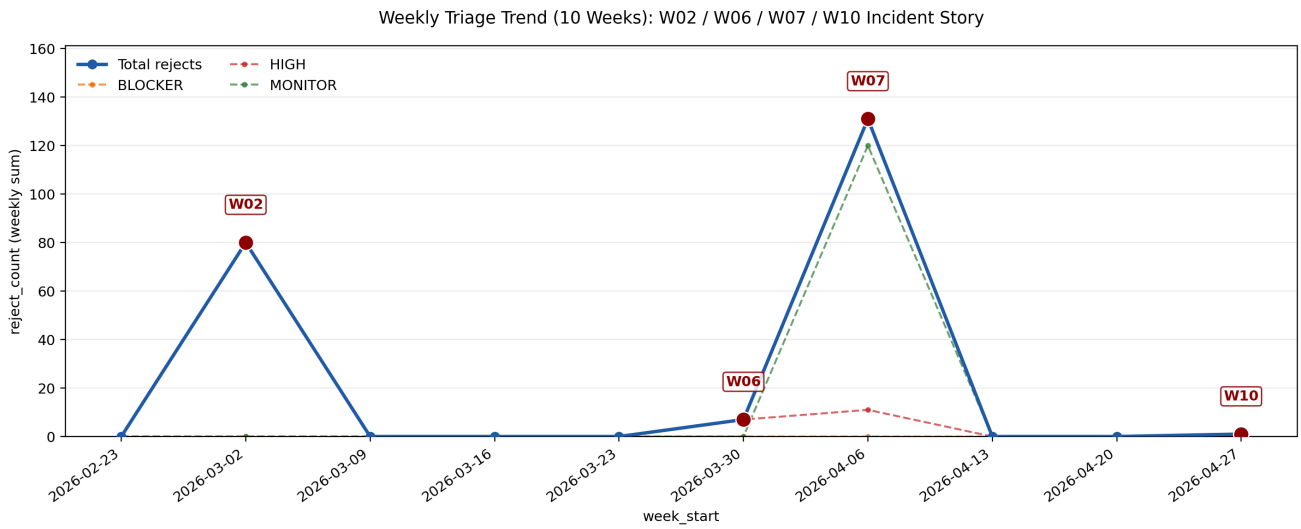
- Total 10-week triage reject volume: **219** (**BLOCKER=80**, **HIGH=18**, **MONITOR=121**).
- Stable weeks are **CLEAN** (0 rejects): 2026-02-23, 2026-03-09, 2026-03-16, 2026-03-23, 2026-04-13, 2026-04-20.
- Four incidents are explicit in trend and storyboard: **W02 blockers**, **W06 eligibility + R902**, **W07 duplicates + R901 + monitors**, **W10 volume + R903 only**.
- Batch anomaly signals detected in-window: **R901=1**, **R902=1**, **R903=1**.

Story at a glance

	week_start	story_label	decision	blockers	high	monitor	top_driver_codes
0	2026-02-23	CLEAN	Monitor	0	0	0	(none)
1	2026-03-02	W02 BLOCKERS	Hold	80	0	0	FUTURE_SERVICE_DATE, NPI_BAD_LENGTH, NPI_NOT_FOUND
2	2026-03-09	CLEAN	Monitor	0	0	0	(none)
3	2026-03-16	CLEAN	Monitor	0	0	0	(none)
4	2026-03-23	CLEAN	Monitor	0	0	0	(none)
5	2026-03-30	W06 R902 ELIGIBILITY-MISMATCH ANOMALY (>2%)	Remediate	0	7	0	SERVICE_OUTSIDE_COVERAGE, ELIG_MISMATCH_GT_2PCT
6	2026-04-06	W07 R901 DUPLICATE-RATE ANOMALY (>1%) + MONITORS	Remediate	0	11	120	ALLOWED_GT_CHARGE, PAID_GT_ALLOWED, NULL_DX
7	2026-04-13	CLEAN	Monitor	0	0	0	(none)
8	2026-04-20	CLEAN	Monitor	0	0	0	(none)
9	2026-04-27	W10 R903 VOLUME-SHIFT ANOMALY (>15%)	Monitor	0	0	1	VOLUME_SHIFT_GT_15PCT




```
Visual QA PASS (Visual 1): bars=12, labels=12, size=142645 bytes
```




```
Visual QA PASS (Visual 2): size=180649 bytes
```

W02 incident: blockers

Triage rollup

	reject_category	severity	reject_count	affected_claims
0	DATES	BLOCKER	20	20
1	PROVIDER	BLOCKER	30	30
2	REQUIRED_FIELD	BLOCKER	30	25

Reject-code detail

	reject_code	rows
0	NULL_MEMBER_ID	10
1	NULL_PROVIDER_NPI	10
2	NPI_BAD_LENGTH	10
3	NPI_NOT_NUMERIC	10
4	NPI_NOT_FOUND	10
5	SERVICE_TO_BEFORE_FROM	10
6	FUTURE_SERVICE_DATE	10
7	NULL_SERVICE_FROM	5
8	NULL_CLAIM_ID	5

W06 incident: eligibility + R902

Triage rollup

	reject_category	severity	reject_count	affected_claims
3	MEMBER_ELIGIBILITY	HIGH	7	7

Reject-code detail

	reject_code	rows
0	SERVICE_OUTSIDE_COVERAGE	6
1	ELIG_MISMATCH_GT_2PCT	1

W07 incident: duplicates + R901 + monitors

Triage rollup

	reject_category	severity	reject_count	affected_claims
5	DUPLICATE	HIGH	11	11
4	CODE_SET	MONITOR	40	40
6	FINANCIAL	MONITOR	80	40

Reject-code detail

	reject_code	rows
0	PAID_GT_ALLOWED	40
1	ALLOWED_GT_CHARGE	40
2	NULL_PROC	20
3	NULL_DX	20
4	DUP_CLAIM_KEY	10
5	DUP_RATE_GT_1PCT	1

W10 incident: volume + R903 only

Triage rollup

	reject_category	severity	reject_count	affected_claims
7	VOLUME_ANOMALY	MONITOR	1	1

Reject-code detail

	reject_code	rows
0	VOLUME_SHIFT_GT_15PCT	1

Top Drivers by Week (triage_by_code)

	week_start	reject_code	reject_count
0	2026-03-02	FUTURE_SERVICE_DATE	10
1	2026-03-02	NPI_BAD_LENGTH	10
2	2026-03-02	NPI_NOT_FOUND	10
3	2026-03-30	SERVICE_OUTSIDE_COVERAGE	6
4	2026-03-30	ELIG_MISMATCH_GT_2PCT	1
5	2026-04-06	ALLOWED_GT_CHARGE	40
6	2026-04-06	PAID_GT_ALLOWED	40
7	2026-04-06	NULL_DX	20
8	2026-04-27	VOLUME_SHIFT_GT_15PCT	1

```
Wrote docs/STORYBOARD.md
```